ESSC CHEM 103 • Midterm 2 Mock Session

Instructions: Show Lewis diagrams, equations and calculation work. Use the supplied data sheet. **This version is designed for timing and stress-management practice, not as a prediction of the actual exam.**

1. Draw Lewis structures with all lone pairs and formal charges for (a) NO_2^+, (b) ClO_3^-, (c) IF_5, and (d) CO. State whether an expanded central-atom octet is used.
2. Draw all equivalent major resonance contributors of NO_3^- and of O_3^-. Label nonzero formal charges; identify what resonance says about measured bond lengths.
3. For XeF_4, and ClF_3, provide electron-domain geometry, molecular geometry, central-atom lone pairs, and whether the molecule is polar. Justify one polarity prediction.
4. Draw structures and determine hybridization at every carbon plus total sigma and pi bonds in (a) CH_3CN, (b) CH_2O, and (c) C_2H_2
5. For acetamide, $CH_3C(=O)NH_2$, identify the hybridization and local shape at each carbon and nitrogen in a basic bonding model. Explain why the amide nitrogen is more nearly planar than an ordinary amine nitrogen.
6. The average C-C bond energy is 347 kJ mol-1 and C=C is 614 kJ mol-1. Select the best C-C estimate from 347, 614, 839, and 1041 kJ mol-1. Explain why bond energies cannot simply be added bond-for-bond.
7. A) For N_2 and N_2^+, calculate bond orders, rank bond strengths and identify which are paramagnetic. B) Explain why Be_2 is not predicted to have a stable bond by the simple valence MO model.
8. Classify HSO_4^-, NH_4^+, CO_3^{2-}, and H_2O as Brønsted acid, base, or amphiprotic; show representative proton-transfer equations (two for each amphiprotic species.
9. Explain and show equations for NH_3 functioning as (a) a Brønsted base in water and (b) a Lewis base with BF_3. Label electron-pair donor and acceptor.
10. Phosphorous acid has structure $HP(=O)(OH)_2$. Write its two stepwise ionization equations and explain why it is diprotic even though its formula is H3PO3.
11. A) Balance the complete reaction of PCl_5 with excess water to make phosphoric acid and HCl. B) Balance the reaction of SO_3 with water. C) State whether the resulting aqueous solutions are expected to be acidic and justify from bonding/oxide character.
12. Predict the stronger acid for each pair: (a) $HClO_2$ vs $HClO_3$; (b) H_2S vs H_2Se. Explain using structural trends, not memorized numerical Ka values.
13. A) Find sulfur oxidation states in SF_4, SO_3, and H_2S. B) Predict whether MgO or SO_3 gives a basic or acidic oxide upon reaction with water and write balanced equations. Explain the connection to metallic vs nonmetallic characters.